# The theta triplet of $\Gamma(4)$: a nilpotent field, a determinant identity, and a period phase along closed geodesics

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**Abstract**

The squares of the three even theta constants span the weight-one modular forms of $\Gamma(4)$, and Jacobi's identity makes the vector $\rho = (-\theta_2^2, -\theta_4^2, i\theta_3^2)$ isotropic. We record four exact statements built on it. (i) With the three quaternion units placed at the half-periods where $\theta_2$, $\theta_4$ and $\theta_3$ vanish, a field $\Phi_\rho$ on the torus $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ is nilpotent at every point; the proof takes two lines, and the other placements fail at a generic $\tau$. (ii) For a bilinear map $Y : M_2(\mathbb{C}) \times M_2(\mathbb{C}) \to M_3(\mathbb{C})$ built from the quaternion units, $\det Y(S, M) = 4 \det S \det M \operatorname{tr}(\epsilon S M^T)$, and the adjugate splits into two terms, each a determinant times a quadratic form. (iii) The real-analytic function $D = -\theta_3^4 h(\lambda)/(4\pi\eta^6)$, with $\lambda = \theta_2^4/\theta_3^4$ and $h(\lambda) = \lambda \log|\lambda| + (1-\lambda)\log|1-\lambda|$, transforms with weight $-1$ and the multiplier of $\eta^6$. (iv) Hence, along the closed geodesic of a hyperbolic element written as a positive word in the two unit shears $L$ and $R$, the amplitude $D\,(dz/ds)^{-1/2}$ returns after one period multiplied by exactly $i^{\#L-\#R}$. We also compute the average over the modular surface of $f = |D|/\|\rho\|^2$, 0.31500491. Scripts for every computation are public.

## 1 Introduction

The vector $u = (\theta_2^2, \theta_3^2, \theta_4^2)$ of squared theta constants is a basis of the weight-one forms of $\Gamma(4)$; the graded ring of $\Gamma(4)$ is $\mathbb{C}[u]$ modulo Jacobi's identity $\theta_3^4 = \theta_2^4 + \theta_4^4$. The modular group permutes the three components through $\mathrm{SL}(2, \mathbb{F}_2) \cong S_3$, which is the action on the three even theta characteristics of the torus. Modular flavour models use $u$ as their weight-one triplet [5, 3]. This note records exact statements about $u$ that arose in a larger computational project [6], stated here on their own. Computations in that public record are cited by their label, such as B1666.

**Notation.** $\tau \in \mathbb{H}$, $q = e^{i\pi\tau}$, and $\theta_2, \theta_3, \theta_4$ are the theta constants in the nome $q$; $\eta$ is Dedekind's eta function and $\chi_\eta$ its multiplier, a character of the metaplectic group $\mathrm{Mp}_2(\mathbb{Z})$ of order 24. The unit shears are $L = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ $(\tau \mapsto \tau + 1)$ and $R = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ $(\tau \mapsto \tau/(\tau+1))$. The torus is $E_\tau = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$, with Weierstrass function $\wp$ and $e_1 = \wp(\frac{1}{2})$, $e_2 = \wp(\frac{1+\tau}{2})$, $e_3 = \wp(\frac{\tau}{2})$, so that $e_1 - e_2 = \pi^2\theta_4^4$, $e_2 - e_3 = \pi^2\theta_2^4$, $e_1 - e_3 = \pi^2\theta_3^4$ [9]. Set

$$\rho = (\rho_i, \rho_j, \rho_k) = (-\theta_2^2,\ -\theta_4^2,\ i\theta_3^2), \qquad \rho \cdot \rho = \textstyle\sum_r \rho_r^2 = \theta_2^4 + \theta_4^4 - \theta_3^4 = 0.$$

The scale of $\rho$ plays no role below except in Section 4, where $\|\rho\|^2 = \sum_r |\theta_r|^4$ up to a common factor that cancels in $f$.

**The multipliers.** Up to scalars the action of $\mathrm{SL}(2, \mathbb{Z})$ on $u$ is the permutation of the even characteristics just described, with $\theta_2^2(\tau+1) = i\theta_2^2(\tau)$ and $\theta_3^2 \leftrightarrow \theta_4^2$. The genuine lifts of this projective action to $\mathrm{Mp}_2(\mathbb{Z})$ that agree with the multiplier $\rho_u$ of $u$ projectively are $\chi_\eta^k \otimes \rho_u$,

$k \in \mathbb{Z}/24$, because two genuine representations that agree projectively differ by a character and the abelianisation of $\mathrm{Mp}_2(\mathbb{Z})$ is $\mathbb{Z}/24$, generated by $\chi_\eta$. Since $\rho_u(s)^2 = -1$ and $\chi_\eta(s)^2 = e^{-i\pi/2}$, a weight $w$ is compatible with the central element for $\chi_\eta^k \otimes \rho_u$ exactly when

$$w \equiv 1 + \tfrac{k}{2} \pmod 2.$$

A particular lift computed in [6] (B1634) is the case $k = 21$, for which weight $\frac{3}{2}$ is compatible; which $k$ occurs depends on a choice of lifts, and we do not use it below.

## 2 A nilpotent field on the torus

Let $\mathsf{e}_i, \mathsf{e}_j, \mathsf{e}_k \in M_2(\mathbb{C})$ be quaternion units: $\mathsf{e}_r^2 = -1$ and $\mathsf{e}_r\mathsf{e}_s = -\mathsf{e}_s\mathsf{e}_r$ for $r \neq s$. Place them at the half-periods where their theta functions vanish: $z_i = \frac{1}{2}$ (where $\theta_2(\pi z)$ vanishes), $z_j = \frac{\tau}{2}$ ($\theta_4$) and $z_k = \frac{1+\tau}{2}$ ($\theta_3$). For each $r$ let $\hat{f}_r$ be the meromorphic function on a double cover with $\hat{f}_r^{\,2} = \wp(z) - \wp(z_r)$ and residue one at $z = 0$.

**Proposition 1.** *The field $\Phi_\rho = \sum_r \rho_r \hat{f}_r\, \mathsf{e}_r\, dz$ satisfies $\Phi_\rho^2 = 0$ at every point of $E_\tau$, for every $\tau$. Of the six placements of the units at the three half-periods, this is, up to signs, the only one with this property at a generic $\tau$.*

*Proof.* The units anticommute, so the cross terms cancel and $\Phi_\rho^2 = -\sum_r \rho_r^2 \hat{f}_r^{\,2}\, dz^2 = -\big[(\rho\cdot\rho)\,\wp(z) - \sum_r \rho_r^2 \wp(z_r)\big] dz^2$. The first term vanishes by Jacobi's identity. For the second,

$$\textstyle\sum_r \rho_r^2 \wp(z_r) = \theta_2^4 e_1 + \theta_4^4 e_3 - \theta_3^4 e_2 = \theta_2^4(e_1 - e_2) - \theta_4^4(e_2 - e_3) = \pi^2\big(\theta_2^4\theta_4^4 - \theta_4^4\theta_2^4\big) = 0,$$

using $\theta_3^4 = \theta_2^4 + \theta_4^4$. For the other placements the same sum is a non-zero combination of the $e_r$ at a generic $\tau$; the script `nilpotent.py` finds it at least 0.43 in absolute value at three generic values of $\tau$, and vanishing to $10^{-30}$ for the stated placement. □

## 3 A determinant identity

Let $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $(C_1, C_2, C_3) = (\epsilon,\ \mathrm{diag}(i,-i),\ \mathrm{diag}(i,-i)\,\epsilon)$, and define the bilinear map $Y : M_2(\mathbb{C}) \times M_2(\mathbb{C}) \to M_3(\mathbb{C})$ by

$$Y(S,M)_{ab} = \sum_{m,n} \big(C_a^T \epsilon\, S\, C_b\big)_{mn} M_{mn}.$$

For $M = \left(\begin{smallmatrix} m_0 & m_1 \\ m_2 & m_3 \end{smallmatrix}\right)$ let

$$A(M) = \begin{pmatrix} m_0^2 + m_1^2 + m_2^2 + m_3^2 & 2i(m_0m_2 + m_1m_3) & -i(m_0^2 + m_1^2 - m_2^2 - m_3^2) \\ 2i(m_0m_1 + m_2m_3) & -4m_1m_2 & 2(m_0m_1 - m_2m_3) \\ -i(m_0^2 - m_1^2 + m_2^2 - m_3^2) & 2(m_0m_2 - m_1m_3) & -m_0^2 + m_1^2 + m_2^2 - m_3^2 \end{pmatrix}, \qquad P = \mathrm{diag}(1,-1,-1$$

**Theorem 2.** *For all $S, M \in M_2(\mathbb{C})$,*

$$\det Y(S,M) = 4 \det S\, \det M\, \mathrm{tr}(\epsilon S M^T), \qquad \mathrm{adj}\, Y(S,M) = \det S\, A(M) + \det M\, P\, A(S).$$

*With $N = M\epsilon$, the map is symmetric, $Y(S, N\epsilon^{-1}) = Y(N, S\epsilon^{-1})$, and $\mathrm{tr}(\epsilon S M^T) = -\big(\det(S+N) - \det S - \det N\big)$ is minus the polar form of the determinant.*

*Proof.* All four statements are polynomial identities in the eight entries of $S$ and $M$. The script `joint_law.py` expands both sides symbolically and finds every residual zero. As a control, replacing $P$ by the identity, or the factor 4 by 3, breaks the corresponding identity. We have not found the identity in the literature; it may be a known identity of the quaternion algebra in disguise. □

**Corollary 3.** *(1) $Y(S,M)$ is singular exactly when $\det S = 0$, $\det M = 0$ or $\operatorname{tr}(\epsilon S M^T) = 0$. (2) The squares $x$ of the singular values of $Y$ are the roots of $x^3 - e_1 x^2 + e_2 x - e_3$ with $e_1 = \|Y\|^2$, $e_2 = \|\det S\, A(M) + \det M\, PA(S)\|^2$ and $e_3 = 16|\det S \det M \operatorname{tr}(\epsilon S M^T)|^2$.*

*Proof.* Both follow from the theorem, since $e_3 = |\det Y|^2$ and $e_2 = \|\operatorname{adj} Y\|^2$ (Frobenius norm) is the sum of the products of pairs of squared singular values. □

*Remark* 4 (an observation without proof). On the diagonal $N = S$ the singular values are in geometric progression, $\sigma_1\sigma_3 = \sigma_2^2$; equivalently, by Theorem 2, $\sigma_2^3 = |\det Y|$. The script `joint_law.py` checks this on twenty random $S$ (numpy generator seed 7) to $10^{-14}$. We do not have a proof.

## 4 A form of weight $-1$

Let $\lambda = \theta_2^4/\theta_3^4$, the modular lambda function, $h(\lambda) = \lambda \log|\lambda| + (1-\lambda)\log|1-\lambda|$, and

$$D(\tau) = -\frac{\theta_3^4\, h(\lambda)}{4\pi\, \eta^6}.$$

It arises in [6] (B1654, B1657) from the norms of Szegő kernels on the punctured torus, but the definition here stands alone.

**Lemma 5.** *$D$ has weight $-1$ and the multiplier $\chi$ of $\eta^6$: $D(g\tau) = \chi(g)(c\tau + d)^{-1} D(\tau)$ for $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$, with $\chi(L) = i$ and $\chi(R) = -i$.*

*Proof.* Under $\tau \mapsto \tau+1$, $\theta_3 \leftrightarrow \theta_4$ and $\theta_2 \mapsto e^{i\pi/4}\theta_2$, so $\lambda \mapsto \lambda/(\lambda-1)$ and $\theta_3^4 \mapsto \theta_4^4 = \theta_3^4(1-\lambda)$. A short computation gives $h(\lambda/(\lambda-1)) = -h(\lambda)/(1-\lambda)$, and $\eta^6 \mapsto i\eta^6$. Hence $D(\tau+1) = iD(\tau)$. Under $\tau \mapsto -1/\tau$, $\theta_3 \mapsto \sqrt{-i\tau}\,\theta_3$ and $\theta_2 \leftrightarrow \theta_4$ with the same factor, so $\lambda \mapsto 1-\lambda$, $\theta_3^4 \mapsto -\tau^2\theta_3^4$, $h(1-\lambda) = h(\lambda)$, and $\eta^6 \mapsto (-i\tau)^3\eta^6 = i\tau^3\eta^6$. Hence $D(-1/\tau) = i\tau^{-1}D(\tau)$. Both sides of the law compose by the cocycle rule and the two maps generate the group, so the law holds throughout, with $\chi$ well defined because $D$ is not identically zero; as a consistency check, $\chi(S_0) = \chi(L) = i$ and $(S_0L)^3 = -I$ give $\chi(-I) = i^6 = -1 = i^2 = \chi(S_0^2)$. Then $\chi(L) = i$, and $R = S_0 L^{-1} S_0^{-1}$ with $S_0 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ gives $\chi(R) = \chi(L)^{-1} = -i$. □

$D$ is real-analytic and not holomorphic, because of $\log|\cdot|$; the law is nonetheless exact. With $\|\rho\|^2 = \sum_r |\theta_r|^4$, the function

$$f = \frac{|D|}{\|\rho\|^2/(4\pi^2|\eta|^6)} = \frac{\pi\,|\theta_3|^4\,|h(\lambda)|}{|\theta_2|^4 + |\theta_3|^4 + |\theta_4|^4}$$

is invariant under $\mathrm{SL}(2,\mathbb{Z})$. It vanishes at $\omega = e^{2\pi i/3}$, where $\lambda = e^{\pm i\pi/3}$ and $h(\lambda) = 0$, and toward the cusp, where $f \sim |q|\,\left|8\pi^2 y + 4\pi(1 + e^{-2\pi i x}) - 32\pi\ln 2\right|$ for $\tau = x + iy$, $y \to \infty$ (the script `cusp_law.py` of [6] checks the leading term to $1.2 \cdot 10^{-6}$ at $y = 5$). Its value at $i$ is 1.0888.

## 5 The period phase

Let $\gamma$ be a hyperbolic element of $\mathrm{SL}(2,\mathbb{Z})$ written as a positive word in $L$ and $R$ (every hyperbolic conjugacy class has such a representative up to sign [8]), and let $z(s)$, $s \in \mathbb{R}$, be its axis in $\mathbb{H}$ parametrised by hyperbolic arclength from the repelling to the attracting fixed point, so that $z(s+\ell) = \gamma z(s)$ with $\ell$ the translation length. Define the amplitude $\psi(s) = D(z(s))\,(dz/ds)^{-1/2}$, the square root continued along the axis.

**Theorem 6.** *$\psi(s+\ell) = i^{\#L - \#R}\,\psi(s)$ for every $s$, where $\#L$ and $\#R$ count the letters of the word.*

*Proof.* Write $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Since $z(s+\ell) = \gamma z(s)$, $dz/ds$ at $s+\ell$ is $(cz+d)^{-2}$ times its value at $s$, and $D(z(s+\ell)) = \chi(\gamma)(cz+d)^{-1}D(z(s))$ by Lemma 5. Let $r(s)$ be the ratio of the continued square root $(dz/ds)^{-1/2}$ at $s+\ell$ to its value at $s$. Then $r^2 = (cz+d)^2$, and since $cz+d$ has no zero on $\mathbb{H}$ and $r$ is continuous, $r = \epsilon\,(cz+d)$ with a single sign $\epsilon$. As $s \to -\infty$ both $z(s)$ and $z(s+\ell)$ tend to the repelling fixed point $x$ and the tangent directions at the two points agree; for $s$ negative enough the argument of $dz/ds$ varies by less than $\pi$ along $[s, s+\ell]$, so the continued square root takes the principal branch of a ratio that tends to a positive number. Meanwhile $cz+d \to cx+d$, the smaller eigenvalue of $\gamma$ (at the repelling point $\gamma'(x) = (cx+d)^{-2} > 1$), which is positive because a positive word has trace greater than 2. Hence $\epsilon = +1$ and $\psi(s+\ell) = \chi(\gamma)\psi(s)$, with $\chi(\gamma) = \chi(L)^{\#L}\chi(R)^{\#R} = i^{\#L-\#R}$ because $\chi$ is a character. □

**Corollary 7.** *The integral of $\psi$ over $n$ consecutive periods is $\sum_{k<n} i^{k(\#L-\#R)}$ times the one-period integral. So its average over $n$ periods tends to 0 as $n \to \infty$ unless $i^{\#L-\#R} = 1$, that is $\#L \equiv \#R \pmod 4$. (The one-period integral itself is in general not zero.)*

The phase is a class function: $\chi$ is a character, so every positive word for the same conjugacy class gives the same $i^{\#L-\#R}$. If the axis passes through a point of the orbit of $\omega$, where $D$ vanishes, $\psi$ has zeros there; the theorem is unaffected.

The script `period_phase.py` checks the theorem on eight words, at three starting points each, to $10^{-28}$. On all 30 674 closed geodesics with trace at most 600, a least-squares phase over two periods returns $i^{\#L-\#R}$ to $1.1 \cdot 10^{-7}$ [6] (B1666).

## 6 The surface average and the trace classes

The average of $f$ over the modular surface, $\frac{3}{\pi}\int_{\mathcal{F}} f\,\frac{dx\,dy}{y^2}$ over the standard fundamental domain, is

$$\langle f \rangle = 0.3150049063,$$

the same at three Gauss–Legendre resolutions (`surface_average.py`). By the equidistribution of closed geodesics [1, 4] and of their discriminant layers [2, 7], the average of $f$ along the closed geodesics of a fixed trace $t$ tends to $\langle f \rangle$ as $t$ grows; all closed geodesics of trace $t$ have the same length $2\,\mathrm{arccosh}(t/2)$, so the plain mean of their time averages is also the length-weighted one. For the classes of trace at most 600, the mean over the top quarter of traces (451 to 600) is 0.3144, against 0.3692 for traces at most 20, with the distance falling roughly as $t^{-1/2}$ [6] (B1666). This is a check of a computation against the theorem, not a new result.

## Data, code and the use of AI

The computations of Sections 2–6 have scripts in `papers/P10_theta_triplet_note/checks/` of the public repository `https://github.com/originaxiom/origin-axiom`, and `verify.sh` runs them all. Two sets of numbers are quoted from the record rather than recomputed here: the phase on the 30 674 closed geodesics of trace at most 600, and the trace-class means of Section 6 (both B1666). The computations and the drafting were done with AI assistance (large-language-model sessions working in the repository under the author's direction); the identities were checked symbolically or numerically as stated, and the author is responsible for the text.

## References

[1] Rufus Bowen. The equidistribution of closed geodesics. *Amer. J. Math.*, 94:413–423, 1972.

[2] W. Duke. Hyperbolic distribution problems and half-integral weight Maass forms. *Invent. Math.*, 92:73–90, 1988.

[3] Xiang-Gan Liu, Chang-Yuan Yao, and Gui-Jun Ding. Modular invariant quark and lepton models in double covering of $S_4$ modular group. *Phys. Rev. D*, 103:056013, 2021.

[4] Grigoriy A. Margulis. *On Some Aspects of the Theory of Anosov Systems*. Springer Monographs in Mathematics. Springer, 2004.

[5] P. P. Novichkov, J. T. Penedo, and S. T. Petcov. Double cover of modular $S_4$ for flavour model building. *Nucl. Phys. B*, 963:115301, 2021.

[6] origin-axiom. The origin-axiom record: preregistrations, instruments, outputs and locks. `https://github.com/originaxiom/origin-axiom`, 2026. tagged release and archival snapshot before circulation.

[7] Peter Sarnak. Class numbers of indefinite binary quadratic forms. *J. Number Theory*, 15:229–247, 1982.

[8] Caroline Series. The modular surface and continued fractions. *J. London Math. Soc. (2)*, 31:69–80, 1985.

[9] E. T. Whittaker and G. N. Watson. *A Course of Modern Analysis*. Cambridge University Press, 4th edition, 1927. ch. XXI.